

## TARCOOLA, Australia

WMO: 946550

Lat: 30.705S

Lon: 134.579E

Elev: 124

StdP: 99.85

Time Zone: 9.50 (AUA)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 1.4        | 2.8 | -11.6                         | 1.4 | 24.5 | -9.1 | 1.8 | 24.1 | 11.4                  | 17.0 | 10.4 | 17.4 | 1.8                   | 240  | 0.519 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 15.4                   | 42.1            | 19.5 | 40.2 | 19.1 | 38.3 | 18.4 | 23.4                | 31.0 | 22.0 | 31.5 | 20.8 | 31.6 | 5.6                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.9                            | 16.8 | 25.2 | 19.6 | 14.6 | 24.2 | 17.7 | 12.9 | 24.2 | 70.5          | 30.6 | 64.9 | 31.3 | 60.7 | 31.4 | 36.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 10.8          | 9.6  | 8.7 | DB                         | -1.3 | 45.2               | 1.3 | 1.6                                                | -2.2 | 46.3       | -3.0 | 47.3       | -3.7 | 48.2       | -4.6 | 49.3 | (4) |
| (5)               |      |     | WB                         | -2.1 | 27.1               | 1.4 | 4.7                                                | -3.1 | 30.5       | -3.9 | 33.2       | -4.7 | 35.9       | -5.7 | 39.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 19.8        | 28.0  | 27.1  | 24.1  | 19.7  | 15.3  | 12.0  | 11.8  | 13.4  | 16.8  | 20.4  | 23.4  | 25.7  | (6)  |      |
|                                                                        | DBStd       | 6.85        | 4.52  | 4.38  | 4.14  | 3.54  | 3.12  | 2.55  | 2.70  | 3.14  | 4.08  | 4.48  | 4.62  | 4.55  | (7)  |      |
|                                                                        | HDD10.0     | 23          | 0     | 0     | 0     | 0     | 1     | 8     | 10    | 4     | 1     | 0     | 0     | 0     | (8)  |      |
|                                                                        | HDD18.3     | 799         | 0     | 0     | 3     | 24    | 106   | 192   | 203   | 159   | 77    | 28    | 6     | 1     | (9)  |      |
|                                                                        | CDD10.0     | 3582        | 557   | 478   | 438   | 292   | 164   | 67    | 66    | 108   | 206   | 321   | 401   | 486   | (10) |      |
|                                                                        | CDD18.3     | 1318        | 299   | 245   | 183   | 66    | 11    | 1     | 1     | 4     | 32    | 91    | 157   | 228   | (11) |      |
|                                                                        | CDH23.3     | 18395       | 4376  | 3419  | 2251  | 752   | 111   | 5     | 12    | 79    | 523   | 1319  | 2233  | 3316  | (12) |      |
| (13)                                                                   | CDH26.7     | 10441       | 2783  | 2078  | 1193  | 289   | 22    | 0     | 1     | 16    | 202   | 657   | 1232  | 1968  | (13) |      |
| (14)                                                                   | Wind        | WSAvg       | 4.4   | 5.1   | 4.7   | 4.4   | 3.8   | 3.5   | 3.4   | 3.7   | 4.2   | 4.7   | 5.0   | 5.0   | 5.0  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 172         | 11    | 22    | 8     | 10    | 18    | 14    | 14    | 15    | 19    | 16    | 15    | 13    | (15) |      |
|                                                                        | PrecMax     | 438         | 66    | 110   | 65    | 84    | 136   | 72    | 56    | 82    | 75    | 90    | 92    | 84    | (16) |      |
|                                                                        | PrecMin     | 51          | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) |      |
|                                                                        | PrecStd     | 87          | 14    | 31    | 13    | 17    | 23    | 15    | 13    | 16    | 18    | 20    | 16    | 17    | (18) |      |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 44.6  | 43.5  | 40.0  | 36.2  | 29.3  | 23.9  | 24.9  | 28.8  | 34.9  | 38.7  | 41.6  | 43.6  | (19) |      |
|                                                                        |             | MCWB        | 20.4  | 20.6  | 18.8  | 17.6  | 15.0  | 13.3  | 12.2  | 13.2  | 15.3  | 16.9  | 18.1  | 19.8  | (20) |      |
|                                                                        | 2%          | DB          | 42.3  | 41.3  | 37.8  | 32.2  | 26.2  | 21.1  | 22.1  | 25.5  | 31.7  | 35.7  | 38.5  | 40.8  | (21) |      |
|                                                                        |             | MCWB        | 19.9  | 19.6  | 17.7  | 15.9  | 14.2  | 12.4  | 11.5  | 12.2  | 14.2  | 15.9  | 17.9  | 18.6  | (22) |      |
|                                                                        | 5%          | DB          | 40.2  | 38.9  | 35.5  | 30.0  | 24.0  | 19.4  | 20.1  | 23.1  | 28.8  | 33.2  | 36.1  | 38.5  | (23) |      |
|                                                                        |             | MCWB        | 19.5  | 19.3  | 17.8  | 15.5  | 13.3  | 11.7  | 10.9  | 11.7  | 13.6  | 15.4  | 17.1  | 18.1  | (24) |      |
|                                                                        | 10%         | DB          | 37.9  | 36.6  | 33.0  | 27.6  | 22.0  | 17.8  | 18.2  | 20.9  | 26.3  | 30.4  | 33.6  | 35.9  | (25) |      |
|                                                                        |             | MCWB        | 18.8  | 18.6  | 17.3  | 14.9  | 12.7  | 11.2  | 10.4  | 11.0  | 12.9  | 14.7  | 16.5  | 17.5  | (26) |      |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 24.8  | 24.4  | 23.4  | 20.7  | 16.9  | 16.5  | 14.9  | 15.7  | 18.0  | 21.8  | 23.9  | 27.4  | (27) |      |
|                                                                        |             | MCDB        | 31.3  | 31.3  | 27.0  | 26.4  | 22.5  | 20.3  | 18.3  | 21.3  | 22.8  | 24.2  | 28.3  | 29.0  | (28) |      |
|                                                                        | 2%          | WB          | 22.6  | 22.8  | 21.8  | 18.9  | 15.8  | 14.3  | 13.1  | 13.9  | 15.7  | 18.0  | 20.0  | 22.1  | (29) |      |
|                                                                        |             | MCDB        | 33.5  | 33.6  | 27.8  | 24.0  | 22.2  | 18.0  | 18.6  | 21.8  | 27.1  | 28.6  | 29.4  | 29.9  | (30) |      |
|                                                                        | 5%          | WB          | 21.4  | 21.4  | 20.3  | 17.4  | 14.8  | 13.0  | 12.1  | 12.8  | 14.6  | 16.8  | 18.8  | 20.4  | (31) |      |
|                                                                        |             | MCDB        | 34.0  | 32.5  | 28.3  | 25.9  | 21.3  | 17.0  | 18.2  | 20.5  | 26.4  | 29.1  | 30.1  | 31.5  | (32) |      |
|                                                                        | 10%         | WB          | 20.4  | 20.2  | 18.9  | 16.3  | 13.9  | 12.0  | 11.2  | 11.8  | 13.6  | 15.7  | 17.8  | 19.1  | (33) |      |
|                                                                        |             | MCDB        | 34.0  | 32.1  | 30.2  | 25.0  | 20.0  | 16.3  | 16.9  | 19.2  | 24.6  | 28.1  | 30.5  | 31.8  | (34) |      |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 15.4  | 14.5  | 13.7  | 13.5  | 12.8  | 12.2  | 13.0  | 14.2  | 15.2  | 15.3  | 15.2  | 15.5  | (35) |      |
|                                                                        |             | MCDBR       | 17.5  | 16.9  | 16.7  | 16.9  | 15.0  | 14.2  | 15.5  | 16.9  | 18.8  | 18.8  | 18.0  | 17.6  | (36) |      |
|                                                                        | 5% WB       | MCWBR       | 5.9   | 5.6   | 5.5   | 6.6   | 6.8   | 7.6   | 7.8   | 7.6   | 7.7   | 7.2   | 6.6   | 6.2   | (37) |      |
|                                                                        |             | MCWBR       | 14.3  | 12.8  | 11.5  | 12.1  | 11.8  | 10.7  | 13.1  | 14.3  | 16.1  | 15.4  | 15.2  | 14.4  | (38) |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.356 | 0.353 | 0.334 | 0.325 | 0.306 | 0.291 | 0.294 | 0.307 | 0.336 | 0.350 | 0.356 | 0.346 | (40) |      |
|                                                                        |             | taud        | 2.459 | 2.490 | 2.515 | 2.526 | 2.548 | 2.576 | 2.559 | 2.500 | 2.425 | 2.394 | 2.421 | 2.470 | (41) |      |
|                                                                        | Ebn at Noon |             | 983   | 968   | 953   | 911   | 882   | 875   | 888   | 916   | 935   | 960   | 977   | 997   | (42) |      |
|                                                                        |             | Edh at Noon | 120   | 112   | 103   | 91    | 80    | 73    | 78    | 92    | 111   | 122   | 123   | 119   | (43) |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 8.02  | 7.29  | 6.21  | 4.70  | 3.58  | 3.07  | 3.34  | 4.30  | 5.69  | 6.74  | 7.50  | 8.00  | (44) |      |
|                                                                        | RadStd      |             | 0.43  | 0.41  | 0.40  | 0.25  | 0.23  | 0.15  | 0.14  | 0.19  | 0.30  | 0.39  | 0.30  | 0.28  | (45) |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           |       | N/A     | N/A    | N/A     | N/A   | N/A   | -2.52       | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) |       | +0.32   | N/A    | N/A     | +0.68 | N/A   | N/A         | N/A     | +129    | +82     |

Nomenclature: See separate page